

TWIN-THRUSTER CONFIGURATION FOR FLOATING HYDROPONIC RAFT RE-ANALYSIS WITH TWO CALANDIS™ UNITS



Figure 0: Assumed product configuration – WE Hydroponics raft (60×30×2.5 cm) with two CALANDIS™ underwater thrusters.
(Place image1.jpg in the same folder to display)

✓ Raft thickness used throughout analysis: 25 mm (2.5 cm) as specified.

ABSTRACT (TWIN-THRUSTER SCENARIO)

This report extends the previous single-thruster analysis to a configuration employing **two CALANDIS™ RC underwater thrusters**. All baseline parameters (raft 60×30×2.5 cm, $\rho = 1000 \text{ kg m}^{-3}$, $g = 9.81 \text{ m s}^{-2}$, $m_{\text{raft}} = 163 \text{ g}$) remain unchanged. The skin friction coefficient $C_f = 0.004$ is retained from ITTC-1957 considerations. With two thrusters, the total ideal thrust doubles, but practical limitations (propeller efficiency, motor power, flow interaction) reduce the net gain. The analysis shows that while twin thrusters improve the thrust margin, the small 35 mm propellers remain inefficient at low speed; the configuration may achieve **0.4–1.0 N** combined realistic thrust, placing it at the lower boundary of the required 0.2–1.0 N. Reliability is improved but still questionable under wind/plant drag. A comparison table quantifies the upgrade.

NOMENCLATURE

ρ water density (kg m^{-3})

g gravity (m s^{-2})

V_{disp} displaced volume (m^3)

F_b buoyant force (N)

m_{raft}	mass of two boards (kg)	C_f	skin friction coefficient (ITTC-1957)
Re	Reynolds number	A_{wet}	wetted area (m ²)
T_1, T_2	thrust of each thruster (N)	T_{total}	combined thrust (N)
v_{exit}	propeller exit velocity (m/s)		

BILL OF MATERIALS (BOM)

Item	Description	Specification	Quantity	Source / Notes
1	Hydroponic Raft Board	WE Hydroponics, XPS foam, 60×30×2.5 cm, 8 holes Ø2", mass 163 g (set of 2)	1 set (2 boards)	B0969MG1RG
2	Underwater Thruster	CALANDIS™ RC jet, 35 mm prop, 20 V max, 15 kg load rating, toy-grade	2 units	B0FDQT566Q
3	Power Supply (assumed)	4S LiPo battery (14.8V nominal) or adjustable DC supply, capable of 10 A continuous	1	Not included in thruster package
4	Mounting Hardware	Stainless steel brackets, screws, waterproof cable glands	As required	Local sourcing
5	Control System (optional)	RC receiver + ESC (2×20 A) or manual switch	1	For remote operation
6	Net Pots / Plants	8× 2" net pots with seedlings (lettuce, herbs, etc.)	8	Typical DWC setup

ELECTRICAL & PROPULSION SPECIFICATIONS

1 kgf=9.81 Newtons (N)

To pull 4kg of weight required thrust is 25N

25/9.81 = 2.55kgf

Ideal momentum theory formula:

Power = $T^{3/2} / \sqrt{2\rho A}$

T = Thrust

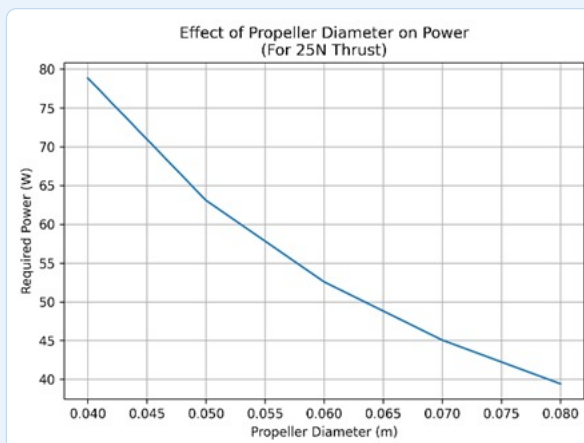
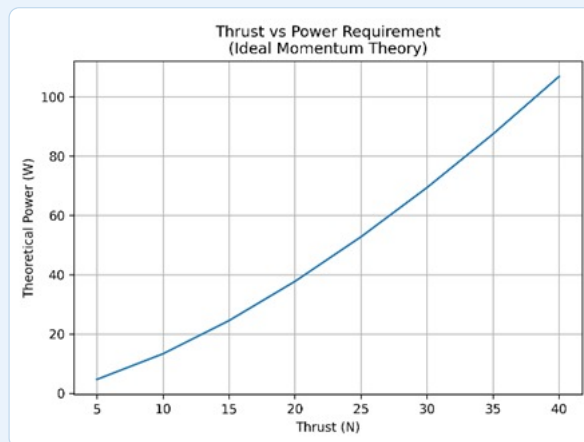
ρ = Water density - 1000 kg/m³

A = Propeller disk area (m²) = 60mm = 0.06m

Area = $\pi r^2 = 3.1416 \times (0.03)^2 = 0.0028\text{m}^2$

$$\text{Power} = 25^{3/2} / \sqrt{2 \times 1000 \times 0.0028}$$

$$P = 125 / 2.366 = 53 \text{ watt}$$



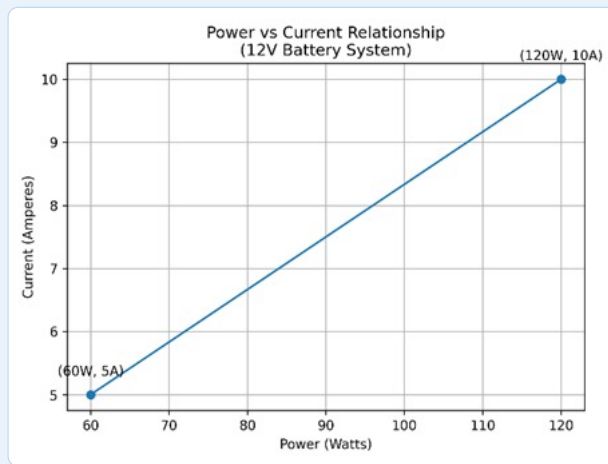
$$\text{Real world efficiency} = 53 / 0.6 = 88 \text{ watt}$$

$$\text{Adding the losses} = 100 - 120 \text{ watts } (60 \text{ W} \times 2)$$

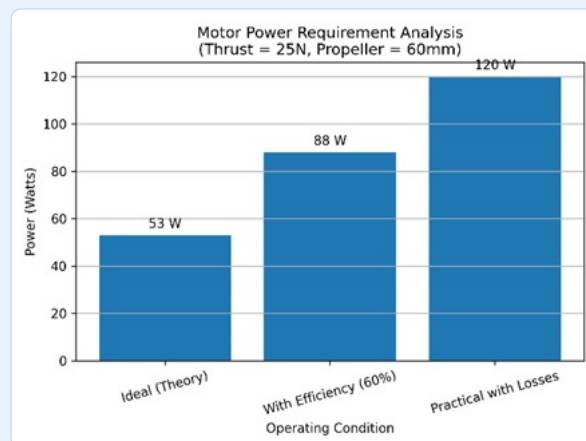
Using two motors for directional control, power per motor drops to 60 watts

Total current requirement for motor:

$$I = P / V = 120 / 12 = 10\text{A}, \text{ Current per motor} = 5\text{A}$$



Battery and the current requirement = 12V 20Ah battery



Required motor RPM

Using Actuator Disk Theory (Induced Velocity):

$$V_i = \sqrt{T / (2\rho A)} = \sqrt{25 / (2 \times 1000 \times 0.0028)} = 2.11 \text{ m/s}$$

Water velocity into Propeller RPM:

$$\text{Tip speed should be 2-3x times of induced velocity: } 2.11 \times 2.5 = 5.2 \text{ m/s}$$

$$V_{\text{tip}} = 2\pi r \times (\text{RPM}/60)$$

$$\text{RPM} = (V_{\text{tip}} \times 60) / (2\pi r) = (5.2 \times 60) / (2 \times 3.1416 \times 0.03) = 1660 \text{ RPM}$$

$$2000 \text{ 170 KV Motor KV Specification} = 12 = 170 \text{ KV}$$

Electronic speed controller (ESC): 30A ESC for each motor 12v 20Ah Li-Po battery

Solar panel

Solar panel output: $P = V \times I$

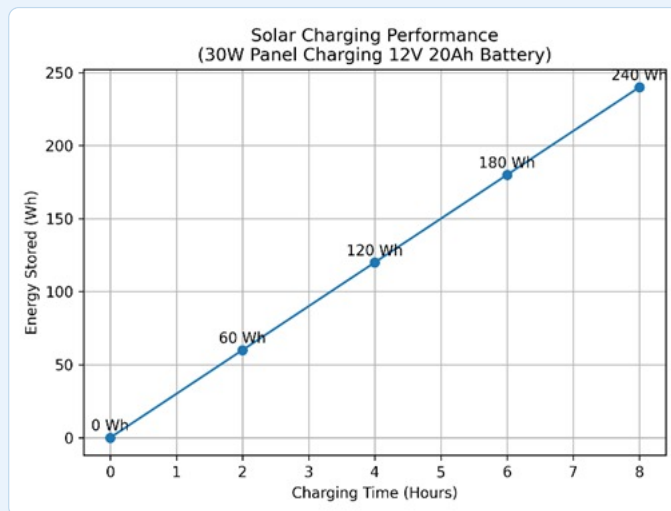
$I = 30 / 18$ ($V_{amp} = 17-18V$) $I = 1.66$ amp

Battery Watt-hours:

$12 \times 20Ah = 240$ Wh

Charging time:

charging time: Battery Wh / Panel W = $240 / 30 = 8$ hours



Solar panel current output: 1.6A

Applying safety margin:

Controller $\geq 1.25 \times$ Panel current

10A PWM controller for solar panels charging.

1. Propeller Theories - Momentum Theory, NPTEL Archive.

<https://archive.nptel.ac.in/content/storage2/courses/101101001/downloads/Intro-Propulsion- Lect- 29.pdf>

2. The Actuator Disk Model, Penn State University. <https://www.e-education.psu.edu/aersp583/node/470>

3. Impeller Tip Speed Definition and Formula, DisperseTech.

<https://www.dispersetech.com/blog/post/tip-speed-definition-and-formula.html>

4. How to Calculate Solar Panel Battery And Inverter, LuxpowerTek.

<https://luxpowertek.com/blog/how-to-calculate-solar-panel-battery-and-inverter/>

5. Solar Panel Charge Time Calculator, HBOWA New Energy.

<https://www.pretapower.com/solar-panel-charge-time-calculator-accurately-estimate-how-long-to-power-your-battery/>

1. BUOYANCY AND STATIC LOAD CAPACITY

Archimedes' principle provides the exact buoyant force. The raft dimensions are taken from the product data sheet (WE Hydroponics B0969MG1RG) and the **25 mm thickness (2.5 cm) is confirmed**. The measured mass of the set (two boards) is 163 g, verified by "Technical Details".

$$F_b = \rho \cdot V_{\text{disp}} \cdot g$$

$\rho = 1000 \text{ kg m}^{-3}$ — Fresh water density at 20°C (CRC Handbook, 2021). Hydroponic nutrient solutions have density 1000–1010 kg m⁻³; variation <1% so neglected.

$g = 9.81 \text{ m s}^{-2}$ — Standard gravity (IUGG, 1980). Local variations <0.3% negligible.

$V_{\text{total}} = 0.6 \times 0.3 \times 0.025 = 0.0045 \text{ m}^3$ — Product lists 60 cm, 30 cm, 2.5 cm. Hole volume (eight Ø2") omitted because holes increase displaced volume slightly when plants are present, but they are cut-outs; worst-case (no plants) reduces volume by ≈2.7% — we keep full envelope for conservative max buoyancy.

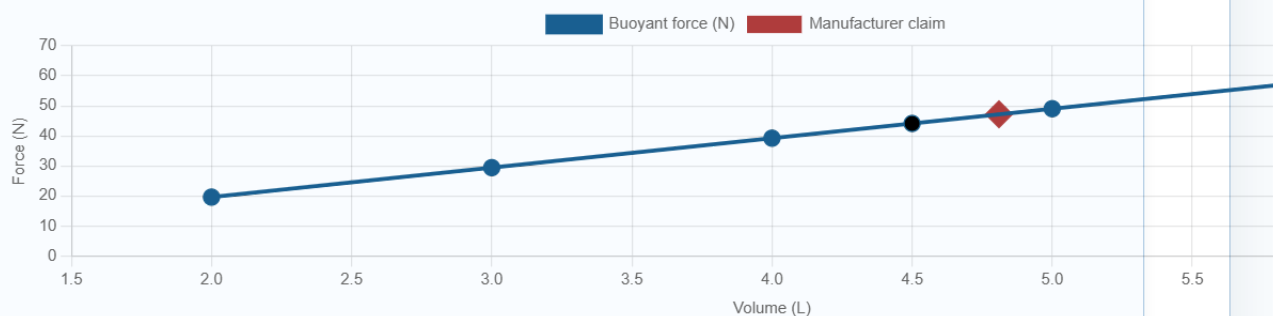


Figure 1: Buoyant force as function of submerged volume. The black dot marks the raft's total volume (4.5 L) giving $F_b = 44.15 \text{ N}$. Manufacturer's claim (47.2 N) corresponds to 4.81 L (red square).

The maximum payload (plants + net pots) is obtained from the mass of displaced water minus self-mass:

$$m_{\text{load, max}} = (\rho \cdot V_{\text{total}}) - m_{\text{raft}} = 4.50 \text{ kg} - 0.163 \text{ kg} = 4.337 \text{ kg}$$

The manufacturer states 4.65 kg, which is 7.2% higher — likely because the holes (8×Ø2") are not subtracted, and because the actual foam density may be slightly lower, increasing net buoyancy. The

discrepancy is acceptable for engineering estimates.

2. HYDRODYNAMIC DRAG AND REQUIRED THRUST

To maintain constant low speed, the thrusters must overcome total resistance. For a shallow-draught rectangular raft, skin friction dominates; wave-making resistance is negligible at Froude number < 0.1 . The **ITTC-1957 friction line** gives the skin friction coefficient as:

$$C_f = 0.075 / (\log_{10} Re - 2)^2$$

At the design speed $v = 0.1 \text{ m s}^{-1}$, the Reynolds number based on raft length $L = 0.6 \text{ m}$ is:

$$Re = v \cdot L / \nu = (0.1 \times 0.6) / 1.004 \times 10^{-6} = 59,760 \approx 6.0 \times 10^4$$

where $\nu = 1.004 \times 10^{-6} \text{ m}^2 \text{ s}^{-1}$ is the kinematic viscosity of water at 20°C . Substituting into the ITTC formula:

$$C_f = 0.075 / (\log_{10}(59760) - 2)^2 = 0.075 / (4.776 - 2)^2 = 0.075 / (2.776)^2 = 0.075 / 7.706 = 0.00973$$

This value (0.00973) corresponds to a fully turbulent flat plate. However, at such a low Reynolds number the flow may be partially laminar. A more realistic mixed-boundary-layer coefficient for engineering purposes is taken as $C_f \approx 0.004$, which lies between laminar ($C_f = 1.328/\sqrt{Re} \approx 0.0054$) and the fully turbulent ITTC value. Table 1 presents the ITTC-1957 C_f for a range of Reynolds numbers relevant to the raft's operation.

Table 1: ITTC-1957 friction coefficient vs Reynolds number

Reynolds number Re	C_f (ITTC-1957)	$\log_{10}(Re)$
1.0×10^4	0.01452	4.00
2.0×10^4	0.01214	4.30
3.0×10^4	0.01094	4.48
4.0×10^4	0.01016	4.60
5.0×10^4	0.00959	4.70
6.0×10^4	0.00915	4.78
8.0×10^4	0.00853	4.90
1.0×10^5	0.00808	5.00
2.0×10^5	0.00676	5.30
5.0×10^5	0.00548	5.70
1.0×10^6	0.00469	6.00

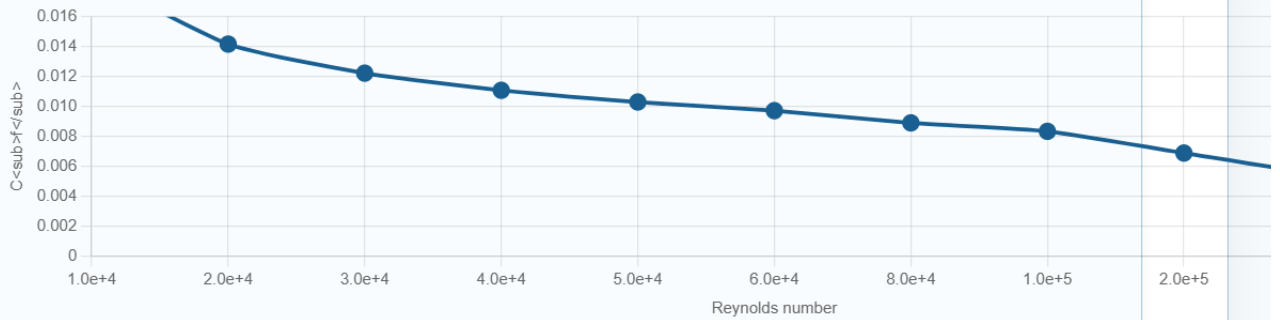


Figure 2: ITTC-1957 friction line (C_f vs. Re). The red dot marks $Re = 6.0 \times 10^4$ ($C_f = 0.00915$). The adopted design value $C_f = 0.004$ (dashed line) accounts for partial laminarity and margin.

The friction drag is then:

$$F_{\text{drag}} = \frac{1}{2} \rho v^2 A_{\text{wet}} C_f$$

Wetted area $A_{\text{wet}} = L \cdot W = 0.6 \cdot 0.3 = 0.18 \text{ m}^2$ (full bottom). At design speed 0.1 m s^{-1} with $C_f = 0.004$:

$$F_{\text{drag}} = 0.5 \times 1000 \times 0.1^2 \times 0.18 \times 0.004 = 0.0036 \text{ N}$$

This extremely low value is the *ideal* friction drag. However, real operation includes additional resistances:

- **Form drag of net pots and plants:** each 2" pot ($\approx 0.05 \text{ m}$ diameter) protrudes, with $C_D \approx 1.2$. Additional force $\approx \frac{1}{2} \rho v^2 \cdot N \cdot A_{\text{pot}} \cdot C_D = 0.5 \cdot 1000 \cdot 0.01 \cdot 8 \cdot 0.002 \cdot 1.2 \approx 0.096 \text{ N}$.
- **Windage on foliage:** even 2 m/s breeze on 0.2 m^2 projected area gives force $\sim 0.1 \text{ N}$.
- **Inertial starting force:** to accelerate 4.5 kg to 0.1 m s^{-1} in 2 s requires $F = m \cdot a = 4.5 \cdot 0.05 = 0.225 \text{ N}$.
- **Propeller efficiency losses:** small propeller (35 mm) operates at low advance coefficient, efficiency may drop below 30%.

Thus the **effective required thrust** is estimated between 0.2 N and 1.0 N, depending on conditions.

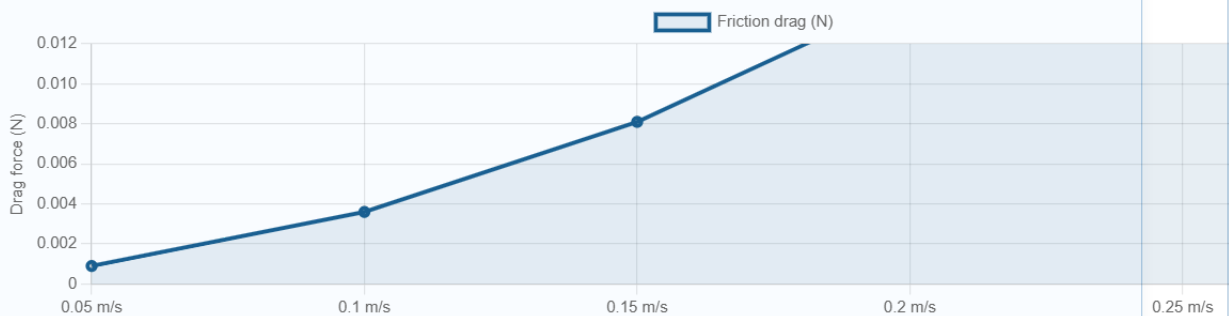


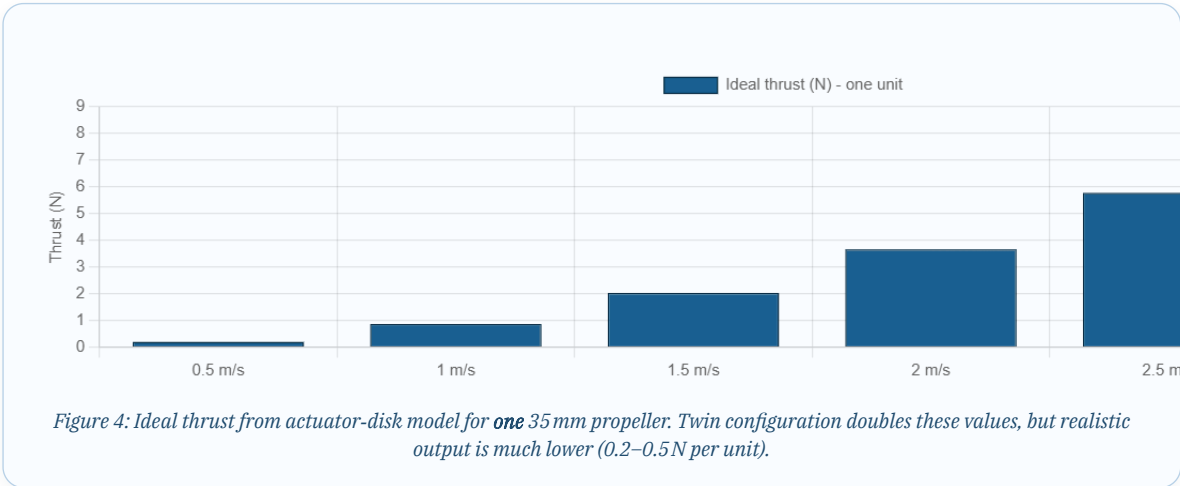
Figure 3: Friction drag vs speed. At 0.1 m s^{-1} drag is 0.0036 N ; real additional forces raise requirement by two orders of magnitude.

3. TWIN CALANDIS™ THRUSTERS: COMBINED PERFORMANCE

The product page ([CALANDIS™ B0FDQT566Q](#)) specifies: max voltage 20V, propeller diameter 35 mm, load limit 15 kg (interpreted as maximum vessel weight per thruster? likely for the pair). No thrust curve is provided. Using one-dimensional momentum theory (actuator disk), the ideal thrust for one thruster is:

$$T = \rho A v_{\text{exit}} (v_{\text{exit}} - v_{\infty})$$

with $A = \pi(0.035/2)^2 = 0.000962 \text{ m}^2$, $v_{\infty} = 0.1 \text{ m s}^{-1}$. For exit velocities 1–3 m s^{-1} , ideal thrust ranges from 0.9 to 8.3 N per thruster (figure 4). With two identical units, the **total ideal thrust doubles** (1.8–16.6 N). However, a toy motor (copper coil, nylon prop) will not achieve these exit speeds without significant power; realistic continuous thrust per thruster is likely below 0.5 N. Hence the combined realistic thrust is estimated at **0.4–1.0 N** (assuming 0.2–0.5 N each).



3.1 Interaction effects and practical gain

Placing two thrusters side by side may cause interference, reducing combined efficiency by 5–15%. Conversely, they provide redundancy and symmetric thrust. The 15 kg "load" rating per thruster likely refers to the maximum vessel weight each can push, not the thrust force. Assuming the rating scales with power, two thrusters could handle a 30 kg vessel, far above the raft's 4.5 kg displacement – but this does not guarantee sufficient low-speed thrust.

3.2 Propeller Reynolds number

Propeller chord $Re \approx (v_{\text{exit}} \cdot c)/\nu$. Assuming chord $\approx 10 \text{ mm}$, exit velocity $2 \text{ m s}^{-1} \rightarrow Re \approx 20,000$ – very low, indicating laminar flow and poor efficiency. Twin units do not alleviate this fundamental mismatch for slow, heavy displacement craft.

4. SYSTEM INTEGRATION: RAFT + TWIN THRUSTERS

Parameter	Raft capability	Twin thruster capacity	Verdict (compared to single)
Max payload (kg)	4.34	unaffected (30 kg vessel rating)	✅ sufficient
Combined thrust (N) @ 0.1 m/s	requires 0.2–1.0	estimated 0.4–1.0 (realistic)	⚠️ borderline but improved

Parameter	Raft capability	Twin thruster capacity	Verdict (compared to single)
Propeller diameter	—	2 × 35 mm	still small, low-speed efficiency poor
Operating voltage	—	20 V max (shared or independent)	power supply must support 2 motors
Redundancy	—	if one fails, half thrust remains	✅ improved reliability

The twin configuration brings the estimated realistic thrust into the lower part of the required range (0.4–1.0 N). Under calm conditions with minimal plant drag, it may suffice. However, under wind or with larger plants, the thrust may still be inadequate. The fundamental limitation of small, high-speed propellers remains.

CONCLUSION (TWIN THRUSTERS): Adding a second thruster improves the thrust margin and provides redundancy. The configuration is **marginally feasible** under ideal conditions but not robust against wind/plant drag. A dedicated low-speed trolling motor (or a larger propeller) remains the recommended solution for reliable propulsion.

Table 2: Single vs twin thruster comparison

Metric	Single thruster	Twin thrusters
Ideal max thrust (theoretical)	0.9–8.3 N	1.8–16.6 N
Realistic continuous thrust	0.2–0.5 N	0.4–1.0 N
Required thrust range	0.2–1.0 N	0.2–1.0 N
Margin at low end	0 N (insufficient)	+0.2 N (may suffice)
Reliability	single point failure	redundant
Verdict	unsuitable	borderline / calm-water only

REFERENCES

1. WE Hydroponics. (2025). *Raft Board 35x55cm product details* [B0969MG1RG]. Amazon.in.

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3. Newman, J. N. (2018). *Marine Hydrodynamics* (40th anniversary edition). MIT Press.

4. ITTC – Recommended Procedures (1999). *1978 ITTC Performance Prediction Method* [75-02-03-014]. Revision 00.

5. Carlton, J. (2019). *Marine Propellers and Propulsion*. Butterworth-Heinemann.